Continuity by Permission: Small-State Agency across Cable Repair, Satellite Licensing and Power Trade

Shivam Gupta
Independent Researcher | India and United Arab Emirates
shivam1720406@gmail.com

Abstract

What makes infrastructure diversification usable during a disruption? This paper argues that small-state agency depends partly on the temporal alignment of permissions, operational readiness and recovery obligations. A structured comparison of nine public documents covers eight institutional units: ASEAN cable-repair coordination, four satellite authorisation regimes, a Kiribati gateway project, and two components of Lao PDR-Thailand-Malaysia-Singapore electricity trade. The documents reveal distinct instruments for reducing administrative friction, retaining regulatory discretion and enabling cross-border services; they do not establish equivalent continuity guarantees. An executable permission-window audit separates availability on a given day from coverage throughout a recovery period. In declared stress scenarios, doubling co-expiring fallback paths leaves temporal coverage unchanged, while staggered permissions or a bounded continuation bridge can improve it. These are conditional accounting results, not estimated outage probabilities. The paper contributes a comparative framework for assessing recovery permissions and proposes an essential-service continuity schedule linking scope, activation, expiry and responsibility. Agency succeeds when governments coordinate the complete recovery path while preserving review authority; it remains limited when improvements govern only one prerequisite or leave service continuation dependent on unresolved decisions.

1. The political problem of usable alternatives

Small and middle powers face a practical infrastructure dilemma. International connectivity and compute require specialised systems that few states can supply domestically. Diversification can reduce exposure to one route or supplier, but an alternative cannot sustain an essential service merely because it exists. Someone must be authorised to activate it, the necessary interfaces must be ready, and permission must remain valid for as long as recovery requires. The political question is therefore how governments turn alternatives into options they can actually exercise.

The Indo-Pacific provides an unusually revealing setting. Cable routes and ownership are increasingly implicated in strategic rivalry [4]. Singapore's Green Data Centre Roadmap anticipates at least 300 MW of additional capacity in the near term, connecting digital growth to energy choices [16]. Pacific governments are introducing satellite permissions with markedly different terms [9-12]. These developments join security, commercial dependence and environmental constraints within the same continuity problem. A government can welcome investment, restrict its scope and preserve discretion, yet still depend on actors beyond its effective direction when disruption occurs.

This paper examines **permission continuity**: whether the authorisations needed for an otherwise feasible fallback cover its activation and the required recovery period. The argument is specific. Small-state agency has a temporal dimension, and infrastructure diversification should be assessed against recovery obligations rather than asset counts alone. Short permissions need not imply weak governance, and long permissions need not imply sovereignty. Their significance depends on scope, readiness, overlapping prerequisites and continuation arrangements.

Three contributions follow. First, the paper compares cable, satellite and power instruments using the same questions about scope, activation and duration. Second, it provides transparent interval accounting that distinguishes point availability from recovery-period coverage. Third, it identifies a policy design problem:

how to preserve review and suspension powers without making essential-service continuity depend on an unplanned renewal during a crisis. The contribution is an institutional comparison with a reproducible diagnostic, rather than a new network algorithm or a claim to have measured national resilience.

2. From network dependence to permission continuity

Weaponized interdependence explains how states with authority over central network nodes can gather information or deny access [1]. It directs attention to the distribution of coercive power. The present question concerns the receiving state's available response: after access is disrupted, what alternatives remain exercisable, by whom, and for how long? Avoiding one chokepoint can move the service onto another jurisdictional or commercial dependency. Fallback therefore requires analysis of the authority governing the replacement path.

Regime complexity supplies a second foundation. Alter and Meunier describe overlapping institutions without a clear hierarchy and examine how their interaction changes cooperation [2]. A recovery operation can depend on actors who did not negotiate the headline infrastructure agreement. A digital regulator cannot necessarily resolve land access, vessel clearance, electricity transit or a provider's commercial commitments. Complexity matters here through the timing and conjunction of decisions, not simply through the number of organisations involved.

The substantive literature already recognises these dependencies. Noor analyses cable repair, cabotage, strategic competition and hyperscaler influence [4]. The CCRC distinguishes slow repair from non-response and examines government options beyond additional cables [5]. Its satellite study treats LEO systems as complementary infrastructure with technical and governance constraints [6]. Interdependent restoration research models joint recovery choices [17], while temporal-network research establishes that timing cannot generally be reduced to static topology [3]. Faster permits, backup satellites and coordinated restoration are consequently established ideas.

The analytical move here is to make permission coverage a comparable object across domains. Administrative speed governs when an action may begin; licence scope governs what action is permitted; duration governs how long it may continue. None substitutes for the others. A recovery permission portfolio consists of the public and contractual authorities needed for a specified service, together with their validity periods and activation conditions. Its adequacy is relative to a recovery requirement, not an abstract label of sovereign infrastructure.

This framing also changes the interpretation of dependence. A state may remain dependent on imported technology while acquiring meaningful authority over continuity conditions. Conversely, local infrastructure can leave critical decisions elsewhere. The relevant distinction is between residual dependence that has a specified response arrangement and residual dependence whose response still requires bargaining under disruption. Permission continuity addresses only part of that distinction; ownership, physical capacity and credible provider performance remain necessary considerations.

3. Research design and evidence boundaries

The study uses a structured documentary comparison with a cutoff of 14 October 2026. Nine focal public documents [7-15] provide eight units: ASEAN coordination, Tonga's event exemption, Niue's trial licence, operating permissions in the Federated States of Micronesia and Sri Lanka, Kiribati's gateway planning, and LTMS importer and wheeling arrangements. Selection seeks variation across institutional forms and reported terms. It is purposive, not a representative sample of states or a census of applicable legislation.

Each unit is coded for document type, authorised scope, reported duration, activation or control arrangements, unreported prerequisites and limits of inference. The evidence ledger supplies locators and source links. A reported term is recorded in its original calendar units; unreported commencement and expiry dates remain unknown. A policy announcement, legal declaration, regulator register and project

disclosure have different evidentiary force and are identified separately. A missing provision in a summary is not treated as proof that no such provision exists elsewhere.

The case comparison establishes institutional design and milestone sequencing. It cannot identify the causal effect of licence duration on outages. Coding is by one researcher, with no intercoder reliability estimate. Public documents generally reveal little about private service commitments or emergency priority. The full ASEAN 2026 guidelines were identified but not retrievable directly; the coded claims rely on Singapore's official summaries [7,8]. Kiribati's disclosure is an August 2025 planning document, not a verified October 2026 commissioning report.

The computational component is deliberately separate. It checks the consequences of explicit, hypothetical permission windows. Named countries are not assigned invented service capacities, renewal probabilities or missing legal terms. Two calendar differences are calculated from documented milestones, but neither is labelled an outage duration. This separation allows a firm conclusion about the logic of permission coverage without presenting a scenario as observed regional performance.

Institutional unit	Reported scope / term	Unresolved assurance question
ASEAN coordination [7,8]	Repair permitting and coordination; no service term coded	Dispatch and domestic implementation
Tonga [9]	Event-specific exemption; six months from Gazette publication	Event eligibility and later permissions
Niue [10]	Spectrum / Internet trial; twelve months	Commencement and continuation terms
FSM [11]	Operating licence; 19 Mar 2024-18 Mar 2044	Other permissions and service commitment
Sri Lanka [12]	Satellite broadband; 12 Aug 2024-11 Aug 2029	Modification and continuity obligations
Kiribati [13]	Interim capacity / IP transit; expiry unreported	Gateway site readiness and commissioning
LTMS importer [14]	Imports; two-year extension to 2026	Exact expiry and subsequent renewal
LTMS wheeling [15]	Lao PDR transit; two years from Jan 2026 signing	Complete contract and permission alignment

Table 1. Documentary comparison. Terms refer to different instruments and are not country resilience scores. Unknowns refer to the reviewed documents, not demonstrated absence of rights. The public ledger supplies locators and document types.

4. Three domains, different forms of agency

4.1 Cable repair: coordinating the right to begin

Singapore's January 2026 statement identifies its initiation of the enhanced ASEAN framework and describes permit streamlining, a single government contact point and coordination among agencies and operators [7]. Its February parliamentary response identifies IMDA as chair of the ASEAN Working Group on Submarine Cables [8]. These are concrete examples of small-state agency through regional agenda-setting and administrative coordination. They can shape practices beyond the state's own waters without requiring ownership of every asset.

Their appropriate outcome measure is narrower than regional resilience in general. Administrative coordination can reduce the decisions needed before mobilisation, but cannot by itself allocate a ship, compatible spares or safe access. The CCRC's repair analysis explicitly distinguishes delay within a functioning commercial system from the absence of response in crisis [5]. The distinction matters politically: a procedure for admitting a vessel is not authority to compel a vessel to arrive. A small state can exercise effective agency over the first while remaining exposed on the second.

The temporal implication is that permit reform and fallback duration interact. A shorter repair process reduces the interval an alternative service must cover. Where mobilisation, transit or non-response

dominate, however, faster administrative processing does not determine the recovery horizon. The paper therefore treats ASEAN coordination as evidence of a relevant intervention, without claiming measured post-2026 repair improvements or inferring a regional dispatch guarantee from endorsement.

4.2 Satellites: authorising alternatives while retaining control

Tonga's 2024 declaration makes an event-specific exemption conditional on notification and regulatory determination, with a six-month term from Gazette publication [9]. It also sets administrative, tax, reporting and payment conditions. Its made date is not established as its commencement date. Crucially, the definition concerns limited significant events; the instrument is not coded as a standing disaster authorisation. This case shows deliberate scope control, rather than evidence that a backup service expired during an outage.

Niue announced a twelve-month spectrum and Internet communications trial in May 2026 [10]. The FSM register lists Starlink's operating permission from 19 March 2024 to 18 March 2044 [11]; Sri Lanka's register lists satellite broadband validity from 12 August 2024 to 11 August 2029 [12]. These terms cannot be ranked directly. They attach to different instruments and may coexist with other permissions. A temporary trial offers a review opportunity; a longer term offers temporal predictability. Neither establishes technical independence, guaranteed capacity or emergency service priority.

Kiribati adds an important activation constraint. Its planning disclosure dates an interim capacity and IP-transit contract to 1 August 2024 and an operating licence to 6 December 2024, a separation of 127 days [13]. It also records rejected siting, an unexploded-ordnance hazard at another location, and continuing lease negotiations for the selected site. These observations distinguish contracting, authorisation and material readiness. They do not establish that all retail satellite service was unavailable, nor that the elapsed interval was caused entirely by regulation.

The comparative result is institutional heterogeneity with a common continuity question. Governments can authorise entry, limit activities and retain decision points, but the usable fallback is the intersection of applicable scope and prerequisites. The CCRC's hybrid-connectivity analysis already recognises terrestrial and provider dependencies [6]. The additional task proposed here is to specify their temporal coverage together, so that the operating permission is not mistaken for the complete recovery arrangement.

4.3 Power trade: assembling permissions across borders

In September 2024, Singapore's EMA announced a two-year extension of Keppel's importer licence to 2026 and an expansion of LTMS trading capacity up to 200 MW [14]. In January 2026, TNB announced a two-year wheeling agreement for up to 100 MW from Lao PDR through Thailand and Malaysia, using existing interconnections, and described full implementation of LTMS-PIP 2.0 [15]. This identifies multiple forms of authority behind a single headline capacity figure.

The announcements are 481 days apart. That is a dated milestone separation, not 481 days without electricity trade. TNB's release separately records a September 2024 Malaysian supply agreement [15]. The evidence supports the narrower conclusion that importer permission, supplier agreement and multilateral wheeling were distinct steps. Their existence and sequencing matter because an importing government's licence cannot unilaterally substitute for the transmission arrangements of its partners.

LTMS illustrates agency through negotiated interdependence. Participating states and utilities construct an arrangement that individual national permissions cannot supply. It also raises a temporal coordination question: how do licence and contractual review periods align when one announced permission reaches 2026 and another two-year agreement begins then? The public record here does not establish a legal gap. It identifies the alignment information a continuity assessment would need, including later renewals and applicable dispatch conditions.

Across the domains, the observed interventions govern different margins: cable procedures address entry into repair; satellite instruments define service scope and review; power arrangements assemble

cross-border transit. Treating these margins as interchangeable evidence of resilience would erase the mechanism through which states exercise agency. Their common feature is that effective action requires multiple permissions to work together over time.

5. An executable audit of the recovery period

Let W(p,k) be the days on which prerequisite k for fallback path p is satisfied. Its effective window is I(p), the intersection of all W(p,k). This notation includes only paths already judged technically feasible, sufficient for the specified essential service and within authorised scope. Unknown prerequisites cannot be silently classified as satisfied. Readiness dates may shorten a window, but legal and operational assessments must establish those dates before the audit is used for procurement.

A disruption starting on integer day t and requiring R days of fallback is covered by one path when every day from t through t+R-1 lies in I(p). Equivalently, [t,t+R) is contained in an effective interval. For a window [a,b), the eligible start days run from a through b-R, when b-a is at least R. A 365-day window therefore covers 365-R+1 candidate start days within [0,365), under the declared integer-day convention.

Two aggregation rules answer different operational questions. Without switching, each path is tested for the full recovery period before eligible starts are combined. With instantaneous, compatible and authorised switching, the effective paths can first be combined and then tested for uninterrupted coverage. The latter is an optimistic bound; it cannot be invoked merely because two providers are licensed. Shared dependencies must be included in each path before either calculation.

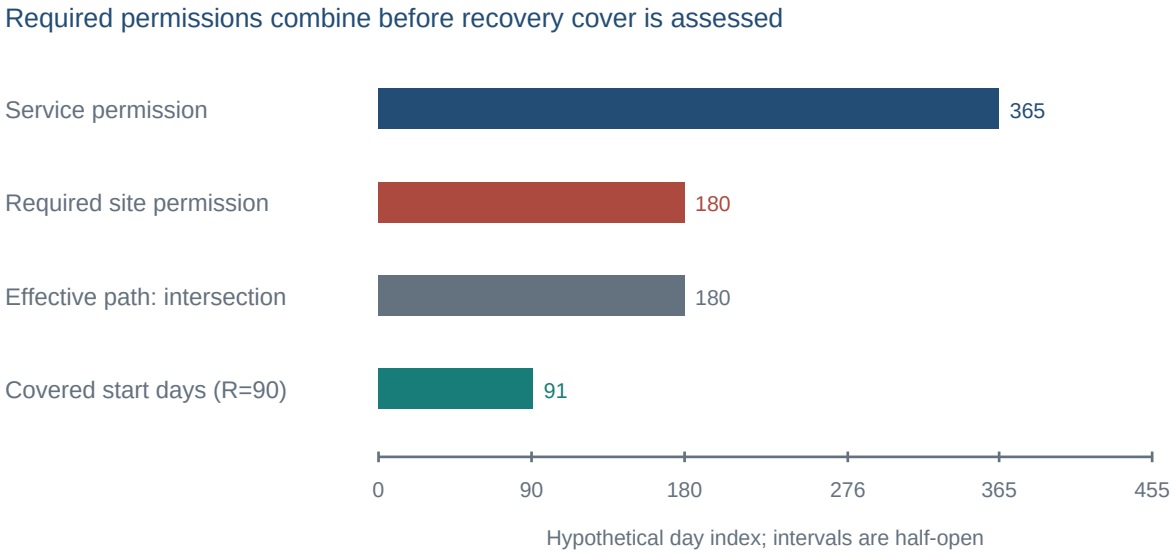

Figure 1. A shorter required permission constrains the complete path. For effective window [0,180) and R=90, start days 0-90 are covered, represented by [0,91). Hypothetical inputs; not a country case.

The resulting conditional proposition is elementary but decision-relevant: additional paths with identical effective windows do not extend temporal coverage. A longer service licence also does not extend coverage beyond an earlier-expiring required site or spectrum permission. Conversely, staggered effective windows can extend coverage if either one path covers each recovery period or a validated handoff connects them. This is ordinary interval logic, not a claimed mathematical discovery. Its value lies in specifying what a governance assessment must establish.

Five portfolios test that logic over 1,800 deterministic configurations: recovery durations of 1-180 days, with and without switching. All candidate starts lie in a 365-day planning horizon, while recovery may extend beyond it. Window lengths of 180 and 365 days are hypothetical inputs, not calendar conversions of the reported six- or twelve-month permissions. No distribution of shock timing is estimated, so the

proportion of covered starts measures temporal coverage, not the probability of successful recovery.

Hypothetical portfolio	Covered starts	Uncovered starts
One path [0,365)	276	89
Two paths, both [0,365)	276	89
Paths [0,365) and [180,545)	365	0
Service [0,365), required site [0,180)	91	274
One complete path continued through day 455	365	0

Table 2. Exact coverage for a 90-day recovery and candidate starts [0,365), without switching. These synthetic inputs are not named-country licence terms. Every path is assumed to have adequate capacity and satisfied scope. Counts are start days, not empirical cases.

At R=90, one path covering [0,365) supports 276 start days; two co-expiring paths support the same 276. Requiring a site window ending at day 180 reduces cover to 91 start days. A second complete path covering [180,545), or extension of the first complete path through day 455, covers all 365 candidate starts. These examples assume adequate service capacity and satisfied prerequisites. They do not price additional provision or establish that any jurisdiction currently grants continuation.

Coverage over a 365-day candidate-start horizon

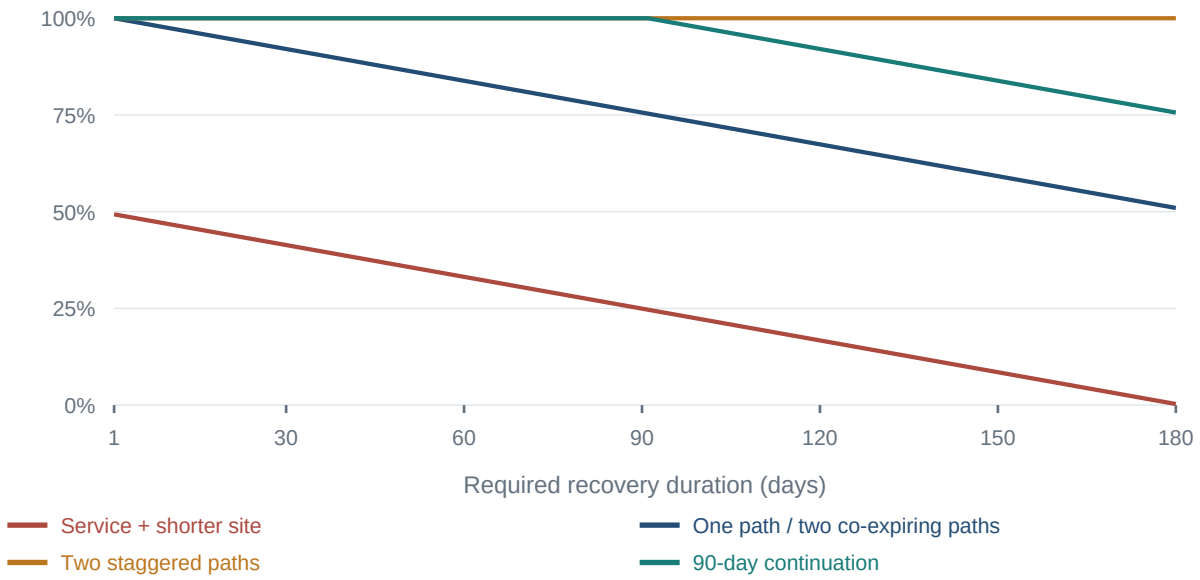

Figure 2. Recovery-period coverage declines as duration requirements increase. One path and two co-expiring paths share the blue curve. Staggering or continuation improves coverage under the specified inputs. No switching is assumed; all values come from the released 1,800-row stress grid.

Switching matters near a different boundary. With windows [0,100) and [90,200), a 30-day recovery has 152 covered starts without switching and 171 under the optimistic handoff rule. The additional nineteen starts require using both paths. The released implementation is checked against an independently written daily oracle on 27,000 small-domain configurations; all agree. This verifies the accounting and endpoint conventions, not the legal coding or the physical effectiveness of a fallback.

The sensitivity analysis reveals a governance trade-off. Extending a complete window can improve temporal coverage, but extending only the most visible permission may do nothing. Staggering permissions

can avoid simultaneous expiry, yet also increase coordination costs or weaken unified review. Adding suppliers can improve protection from provider failure even when temporal coverage is unchanged. The audit isolates one margin and should accompany, rather than replace, capacity and failure-dependence analysis.

6. Governing continuity without surrendering review

The policy implication is an essential-service continuity schedule agreed before disruption. For each specified service, it would identify the complete fallback path, authorised activities, activation responsibility, readiness evidence, validity periods and decisions needed for continuation. It should distinguish the authority to permit an action from the obligation to provide it. Commercial recovery commitments, capacity and applicable public permissions belong in the same schedule because failure at any required prerequisite can defeat the intended option.

A bounded continuation bridge is one possible instrument. Where legally available and agreed in advance, it could preserve an existing essential-service scope for a limited recovery period while a timely review remains pending. It should have a fixed maximum duration, notification and reporting obligations, capacity commitments and an express ability to suspend unsafe or unlawful operation. Administrative continuation has precedents in licensing practice [18]; the proposal is its explicit coordination with a documented recovery requirement, not invention of automatic renewal.

This instrument carries costs. An incumbent could use continuity arguments to resist a justified refusal or weaken competition. The bridge should therefore protect the named service and affected users, not guarantee indefinite market access. A trial licence may intentionally restrict entry while regulators learn about interference, affordability and incumbent-network effects. The relevant design question is how essential services transition if the trial ends, including alternatives to renewing the same supplier. Accountability and continuity must be planned together.

Small and middle powers can exercise agency at three levels. Nationally, they can coordinate regulators and procurement bodies around complete paths. Regionally, they can align repair procedures and negotiate interoperable continuity commitments. Through financing and purchasing, they can require evidence of lawful activation and coverage before treating redundancy as delivered. The ASEAN and LTMS cases demonstrate actual coordination instruments; a cross-domain continuity schedule remains a proposal whose effectiveness would require implementation evidence.

The practical application is an assurance workflow for governments, operators and infrastructure financiers. The software can identify dates on which a declared recovery commitment lacks permission cover and show which prerequisite constrains it. Commercial value would come from validated evidence, maintenance of changing permissions and integration with operational exercises. The interval calculation alone is not a defensible technological moat. Publishing it makes the underlying claims inspectable and prevents a proprietary score from hiding unknowns.

7. Conditions, limitations and conclusion

The comparison answers three questions about regional agency. Strategic rivalry over routes and suppliers, concentrated commercial infrastructure, new satellite entry and energy-intensive digital expansion have made continuity politically consequential [4-6,16]. Small and middle powers can shape it through administrative coordination, scoped entry rules and negotiated cross-border arrangements [7-15]. Those efforts produce usable options when authority, readiness and service commitments cover the required recovery path and period.

They remain incomplete when a reform addresses only one prerequisite, when permission scope excludes the needed activity, or when continuation and provision require unresolved decisions. These are conditional mechanisms supported by the documentary comparison and exact accounting. The study does not

demonstrate that any sampled state experienced an expiry-induced outage, rank their resilience, or show that longer licences cause better outcomes. Recoverability also depends on repair queues, weather, power, affordability and provider performance beyond the model.

Future empirical evaluation should record actual activation dates, renewal decisions, fallback capacity and recovery completion, then test whether identified permission boundaries constrain service. Private agreements and later implementing instruments may alter the picture presented here. The public ledger makes those revisions possible without rewriting a hidden scoring system. The central conclusion nevertheless holds at the level established: asset diversification and point authorisation do not, by themselves, establish recovery-period coverage. Small-state agency includes the ability to align the permissions that make an alternative usable while retaining legitimate authority to review it.

Data and code availability

The evidence ledger, source locators, interval implementation, oracle tests, stress inputs and figure-generation scripts are available at https://github.com/shi1720/permission-continuity. All quantitative results can be reproduced without external datasets or paid services. Documentary observations and hypothetical configurations are stored separately.

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